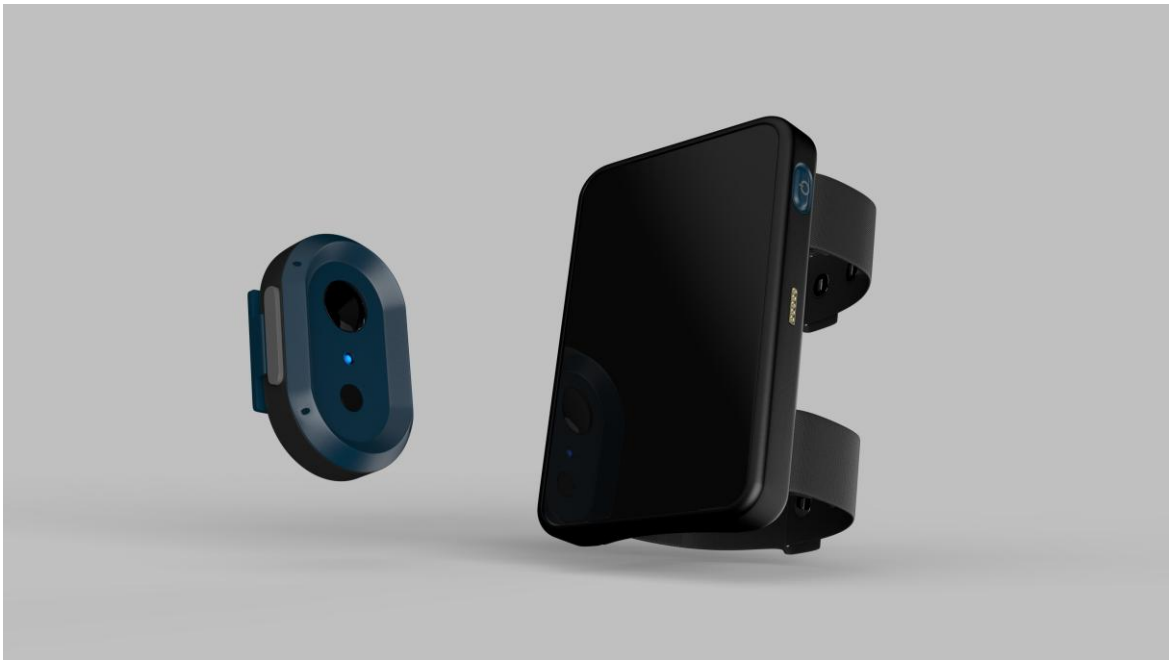
Hamayon Jafari, BFA3
hamayonjafaridesign@gmail.com

UMEÅ UNIVERSITET

Handy Picking System



Hamayon Jafari

Report for thesis project, bachelor's program in Industrial Design.

Examiner: Eva-Lena Bäckström

Advisor to examiner: Per Sihlen

Supervisor from the Umeå Institute of Design: Lars Johansson

External supervisor: Peter

Hamayon Jafari, BFA3
hamayonjafaridesign@gmail.com

UMEÅ UNIVERSITET

Table of contents

1. Abstract ... 3
2. Background and collaboration partner ... 5
3. Design problem ... 5
4. Limitations ... 6
5. Target group ... 6
6. Specified problem statement ... 6
7. Goal and purpose ... 6
8. Design process ... 6
9. Market analysis ... 7
10. Sustainability goals ... 8
11. Research ... 9
    11.1. Surveyed the warehouse ... 11
    11.2. Pitch of the project for the technician warehouse ... 11
    11.3. Eight interviews ... 12
    11.4. Work environment ... 13
    11.5. Activity analysis ... 15
    11.6. System map ... 15
12. Process ... 16
    12.1. Focus area ... 16
    12.2. Function analysis ... 16
    12.3. Brainstorming phase ... 17
    12.4. Overall concept ideas ... 18
    12.5. Selection of concept ... 19
    12.6. Form expression ... 20
    12.7. Details about the chosen concept ... 20
    12.8. Size arm dimensions ... 21
    12.9. Design exploration of the chosen concept ... 21
    12.10. Prototypes ... 22
    12.11. Follow-up to user ... 23
    12.12. Model development ... 24
13. Results ... 24
    13.1. Selection of materials ... 25
    13.2. Scenario ... 25
    13.3. Interface distributed ... 26
    13.4. User scenario with interface ... 29
    13.5. Product name ... 31
    13.6. Renderings ... 31
14. Follow-up on the UN's Global Goals ... 34
15. Discussion ... 35
16. Reference list ... 36
17. Attachments ... 38

All image material not referenced belongs to the author.

1. Abstract

English

This bachelor's thesis in the Industrial Design programme, completed at Umeå Institute of Design, focuses on improving the picking system within food warehouses. The project addresses challenges with the current pick-by-voice system, which often causes errors, communication issues, and reduces social interaction among workers. Through extensive user research including warehouse surveys, eight interviews with users, and observations the key problems were identified, such as voice recognition inaccuracies, user discomfort, and inefficiencies in the workflow. Using the user-centered design approach and the Double Diamond process, the project led to the development of a new concept called "Handy Picking": a wearable device featuring an industrial-grade touchscreen in a compact format and haptic feedback. The second component is an attachable device for the jacket, integrating AI support to assist the picking process. The solution aims to enhance picking accuracy, efficiency, and user satisfaction, while contributing to sustainability goals by improving working conditions and minimizing food waste. Materials, form language, and real-world usability were carefully considered through iterative prototyping and user feedback. The project offers a practical, future-ready alternative to the outdated system, combining reliability, adaptability, and user well-being.

Svenska

Detta kandidatarbete inom programmet för industridesign, genomfört vid Umeå Designhögskolan, fokuserar på att förbättra plocksystemet i livsmedelslager. Projektet tar itu med utmaningar med det nuvarande röststyrda plocksystemet som heter pick-by-voice, som ofta leder till fel, kommunikationsproblem och minskad social interaktion mellan personalen. Genom omfattande användarstudier, inklusive undersökningar i lager, åtta användarintervjuer och observationer, identifierades de främsta problemen, såsom bristande röstigenkänning, användaroduglighet och ineffektivitet i arbetsflödet. Med hjälp av en användarcentrerad designmetod och dubbel diamant-processen utvecklades ett nytt koncept kallat "Handy Picking": en bärbar enhet med en industriell pekskärm i kompakt format och haptisk återkoppling. Den andra komponenten är en fästbar enhet för jackan, som integrerar AI-stöd för att assistera plockprocessen. Lösningen syftar till att öka plocknoggrannheten, effektiviteten och användarnöjdheten, samtidigt som den bidrar till hållbarhetsmål genom att förbättra arbetsvillkor och minska matsvinn. Materialval, formspråk och praktisk användbarhet har noggrant beaktats genom iterativ prototypframtagning och användarfeedback. Projektet erbjuder ett praktiskt, framtidssäkrat alternativ till det föråldrade systemet, med fokus på tillförlitlighet, anpassningsförmåga och användarens välmående.

2. Background and collaborating partner

In the food warehouse industry, the operation system is essential, and the name of the operating system is the warehouse management system (WMS). Everyone has their tasks to do, and people work in different sectors. Picking goods is one of the important aspects because you need to pick the right goods with the right amount and put them in the right roll cage trolley to make sure the customer receives their order. When you pick goods and place them on top of each other, they need to be balanced to prevent anything from falling off the carts.

In the warehouse, trucks transport goods from other food industry sectors, and this transport can be from inside the country and outside the country, and the warehouse workers store these goods in the buffer. There are staff members who drive reach trucks to place goods on shelves, while other staff pick goods from the shelves and bring them to the finalization area with the low-level order picker. The finalization task is to ensure that the goods are correctly sorted and picked for delivery to the right customers. Finally, the goods are delivered to customers by truck.

In the food warehouse, there is also a frozen food sector where staff pick goods in the same way as in the regular sector and bring them to their own finalization sector because the temperature needs to be colder to preserve the food. In my thesis project, I focus on these people who pick goods from the shelves and drive them with the low-level order picker to the finalization area. In my thesis project, there is no collaboration partner who will be involved, but I have external tutoring with a food warehouse in Västerbotten.

3. Design problem

The project explores the problems that people face in the picking section of the warehouse. In the warehouse, these workers do one of the most important tasks ensuring that customers receive their goods correctly. The issue lies with the Warehouse Management System (WMS), which runs the orders and processes everything that customers purchase.

This system is connected to the pick-by-voice system, pick-by-voice is a voice directed picking system. It consists of four parts: a headset dock, a microphone, a battery for the microphone, and a remote. Workers wear headsets and receive spoken instructions to pick items, then confirm actions verbally. While pick-by-voice makes the work more convenient, it has a few disadvantages: it sometimes mishears numbers, makes errors, or does not hear anything at all. The project's starting point is to develop solutions that make the work easier for the workers.

4. Limitations

In a food warehouse, there are a few areas where people work. There is the base sector, the frozen sector, the finalization sector, the elevator platform, and people who drive narrow aisle trucks to fetch goods from the buffer to the base and freeze areas. The project is limited to people who work in the base sector because the bachelor's thesis project is a 10-week project, and there is little time to focus on other areas.

5. Target group

My target group is people who work in food warehouses in Västerbotten, between the ages of 18 to 60 years old. This includes adults who are students working part-time, as well as older and younger individuals who work full-time, both male and female.

6. Specified problem statement

"How might we design a product for food warehouse pick system that streamlines the workflow of warehouse employees while considering efficiency and reliability?" More specifically, it can be divided into three subcategories:

How can employees' complaints be reduced when picking up goods from shelves?

How can a better product be designed to increase accuracy in workers' workflow?

How can a product with better functionality be designed to provide workers with the necessary information?

7. Goal and purpose

This thesis project aims to improve the pick-by-voice system in the food warehouse. The hope is that the results of this thesis project can be a source of inspiration for requirements specifications for the future food warehouse industry.

8. Design process

The design process that will be used in this thesis project is the double diamond. This type of process involves dividing the project into four phases. The first phase is about research, where as many resources as possible are collected within the topic. The second phase focuses on analyzing and drawing conclusions from all the resources that were collected. This is then followed by the ideation phase,

where many ideas and solutions are generated. Finally, the project ends with a concept development phase, where the ideas are tested and refined by sketching, prototyping and modelling.

This thesis project mainly focuses on the user-centered design perspective. It also considered the Sustainable Development Goals of this project. In the initial phase of the thesis project, interviews were held with the warehouse workers, who work in the base pick section. In the beginning, I interviewed four people, and in the third week of the project I interviewed four more people to get more insight. In addition, study visits are also surveyed. I had tutoring with my external partner every week to share the process and get feedback. The research that was previously carried out was analyzed, and after that a function analysis and specific problem formulation has been made. (see figure 1)

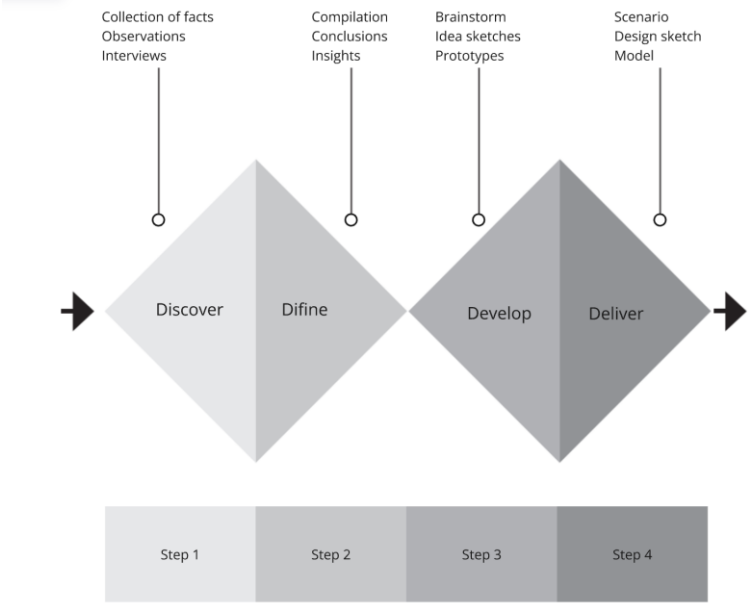


Figure 1: Double diamond. (British Design Council, 2005)

9. Market analysis

The evolution of the pick-by-voice system in logistics is in two generations. The first generation relied on paper, where a list of orders was used for picking. Nowadays, pick-by-voice systems are widely used in logistics, with most of them having similar functionality and usability. There are also other systems out there, such as pick-by-light and wearable computers. These are usually used for piece goods products. Companies that offer pick-by-voice systems include:

Honeywell (Honeywell, n.d.), Zebra Technologies (Zebra Technologies, n.d.), Körber (Körber, n.d.), Dematic (Real Time Logistics, n.d.), Voiteq (Logistics IT, 2014), and Lucas Systems (Lucas Systems, n.d.). This is an analysis of the

differences between paper lists, pick-by-voice, pick-by-light and wearable computers. (see figure 2)

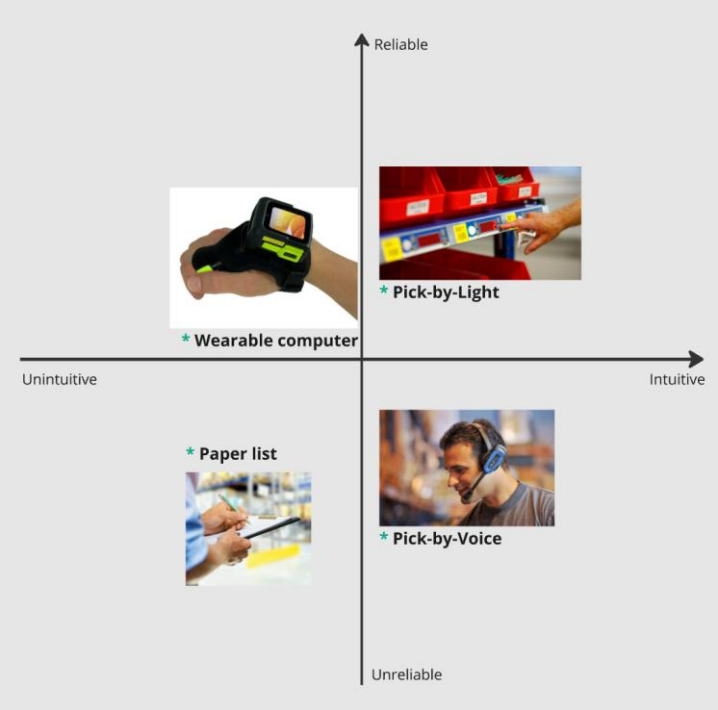


Figure 2: Benchmark.

10. Sustainability goals

Regarding sustainability, the 8th and 9th Sustainable Development Goals are relevant to the warehouse industry because they focus on work and economic growth. Improving working conditions for employees is essential for food warehouses, benefiting both the warehouse and its customers. Goal number 9 can contribute to increasing productivity and minimizing food waste, enhancing the physical environment, such as through better ergonomics and safety standards and ensuring that workers are healthier. Additionally, implementing sustainable packaging solutions for outbound deliveries can further reduce environmental impact. (see figure 3)



Figure 3: Sdgs.un.org (2023).

11. Research

The value of the voice picking solutions market is to grow to 15,3% per year and reach 2,9 billion US dollars by 2025. Pick-by-voice helps workers focus on their tasks, enabling users to use their hands and eyes free from reading directions. They select the right quantity and place it in the appropriate place. Since the worker's hands are free from picking up and putting down devices, their eyes and minds can concentrate on picking the correct goods. Regarding the errors, studies show that the average picking error rate is between 1% and 3% or between 10 to 30 errors per 1000 picks. (Jenkins, A. 2021)

Voice picking requires little training, as it guides the employee through tasks with simple and easy-to-understand voice prompts. Each task that a worker performs is driven step-by-step through voice commands, which means no training on reading print or printing labels is required.

Here is how pick-by-voice works: each warehouse employee assigned to picking duties receives a voice-picking device (see figure 4), which is usually a special-purpose mobile computer linked to a headset with a microphone. The picking system imports customer orders from the warehouse management system (WMS) or enterprise resource planning (ERP) software. These orders are recorded, sequenced, and released for processing. The pick-by-voice device speaks instructions to the warehouse worker, detailing the picking tasks and saying where to go.



Figure 4: (Del Buono, 2015).

The employee can feed information back into the system with voice. When the worker arrives at a specific warehouse location, they verbally provide confirmation digits to verify the location. The pick-by-voice confirms the aisle and provides the operator with the picked goods, the operator verbally confirms the goods. When the task is done, the pick-by-voice sends the picker to the next pick location. All interactions between the picker and the voice application are tracked, which means managers can view the progress of the orders as they are fulfilled. They can also see if they are lagging behind or ahead for today's work time.

The conditions in the frozen and fresh food sector and a specific challenge to picking workers as they go about their daily work, it is necessary to wear gloves, a cap, and warm clothes to protect themselves from the cold temperature. The temperature is down to -23 Celsius. The tough working environment means that warehouses must provide their workers with technologies that they will enjoy working with and will assist them efficiently from an ergonomic perspective.

The first-time testing of smart glasses for picking quantity in daily operations in the Nordic region. The trial was carried out in the dairy in Kalbakken outside Oslo. Roughly a third of the volume in the dairy is picked automatically, and in this warehouse, workers can receive information about the next quantity in a pair of smart glasses. One of those who has tried picking with glasses, Simen Haugland says, *"I find it very effective; it is a slightly different way of working."* There have been concerns that smart glasses could cause eye strain and headaches. But this is something that the experiment could not show. Some advantages of the technology are that the learning time is shorter than with pick-by-voice, and it allows for better adjustment of the picking pace. The glasses used in the experiment are Glass Enterprise Edition (formerly Google glasses). The German company Picavi is responsible for the software and system integration. (Udikas, M. 2019)

The pick-by-voice technology has become easier to use and expand, now logistics facilities no longer need multiple software applications, and everything is integrated into the warehouse management system (WMS). Niels Balvert, logistics manager at Boland, says *"With pick-by-voice, our operators work 10% faster."* (Mecalux, 2021).

The Even Realities G1 smart glasses are designed to seamlessly integrate technology into everyday glasses, having features like real-time language translation, AI assistance, turn-by-turn navigation, and a personal assistant. They have a Holistic Adaptive Optical system, in addition to having a micro-LED projector that displays crisp, green digital text (640 x 200 pixels density) on the lenses, offering a heads-up display (HUD) that appears to float about five feet in front of the user. (Haslam, 2024).

The project's first research has three important elements:

1. Surveyed the warehouse area and identified the problems that workers face in their daily work time.

2. Pitch of the project for the technician warehouse in Västerbotten introduced the project through a brief.
3. interviewed eight workers of different ages and genders at the warehouse to get insight from their experience with the pick-by-voice system.

The expected outcome of the project is a device that is adapted to the user's needs.

11.1. Surveyed the warehouse

Before the thesis project began, the warehouse was surveyed to identify problems in different aisles. When I worked in the warehouse, I also discovered some problems at that time as well. For instance, I noticed that a few people, when starting a new round, forgot to lift the forklift forks. They first placed three boxes on the forks and then started driving the forklift. After moving a few meters forward, they realized that they had left the boxes behind. (see figure 5)

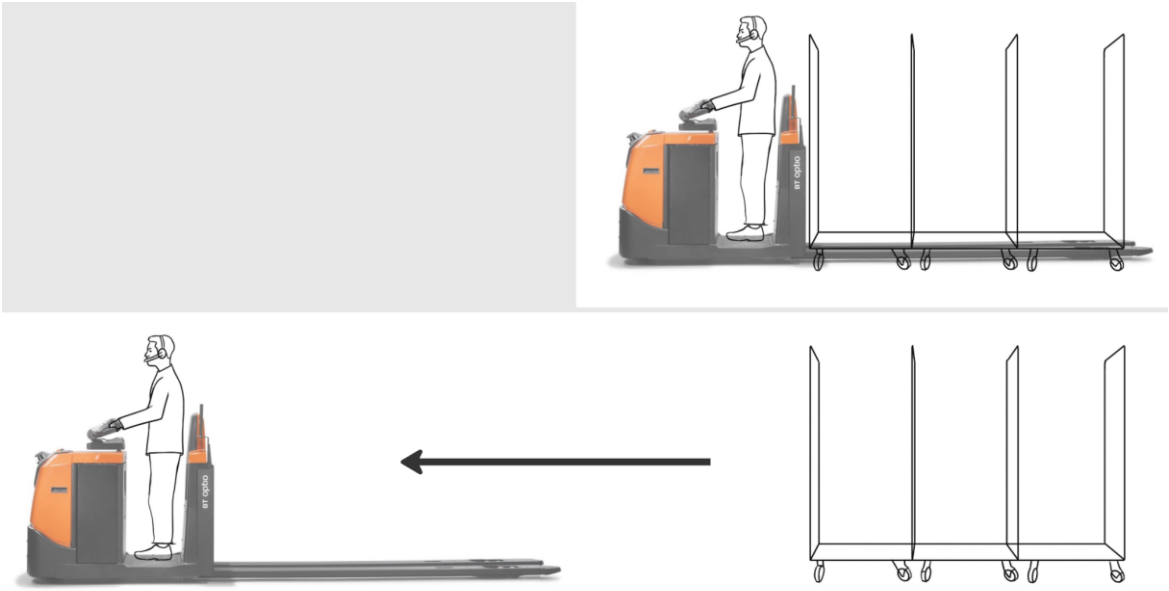


Figure 5: When the worker forgets to lift up the forklift forks.

11.2. Pitch of the project for the technician warehouse

Before the thesis project began, I introduced the project through a brief. Summarized the project to my external tutor he works as a technician staff in the warehouse. It seems interesting to him and now he is my external tutor to guide me through the project.

11.3. Eight interviews

Eight people were interviewed of different ages and genders to gain diverse perspectives and make the project as holistic as possible. I conducted two rounds of interviews because it wasn't possible to interview everyone at once. The first interview was made on 2025-03-04, during which I interviewed four people in a food warehouse in Västerbotten, and the second was completed on 2025-03-17. I prepared ten questions for the interviews; see them under the 'Attachments' heading from page 38 to 42. I did not create a survey to send to people because it wasn't necessary for this project. I needed to ask specific questions, which is why I chose to conduct interviews instead. Below is a summary format of the interviews and two quotes.

One of the users who previously worked with pick-by-light shared his opinion about how the work should be: *"It doesn't work well with a headset; it makes the work stressful. The social part of the job disappeared. I believe work should be enjoyable."*

One of the users explained their experience with pick-by-voice: *"Sometimes, when there's a double row, you take a bucket from level 30, go down to put it in the box, and then you need to fetch another one. Then you have to repeat the same process again, and it takes time."*

The pick-by-voice system demonstrates poor voice recognition capabilities. It frequently disconnects, lags, and exhibits ghosting issues, resulting in error codes that disrupt workflow efficiency. Additionally, there is a lack of flexibility for correcting errors in real time. The system often misinterprets commands and numerical inputs; for instance, "31" it recognized as "91," and "44" heard as "talk man", leading to operational mistakes. One user reported that the headset is uncomfortable and bulky, and that the system sometimes exhibits delayed responses to commands. Furthermore, the system manages quantity and weight inputs poorly, particularly when handling meat and fruit products. Several users expressed a desire for integrated tools such as a clock, calculator, and visual progress tracker to enhance the picking process. Overall, these issues negatively impact workflow, social interaction, and overall user satisfaction.

11.4. Work environment

I created an illustration of the work environment in the base area, which is the focus of my thesis project. In this image, arrows represent the workers walking path with low-level order picker. Each colour represents a different sector of the base area. This is one example of a food warehouse layout; however, another food warehouse in southern Sweden may look different. The focus of this project is to design a new solution that can be implemented in other food warehouses in Sweden. (see figure 6)

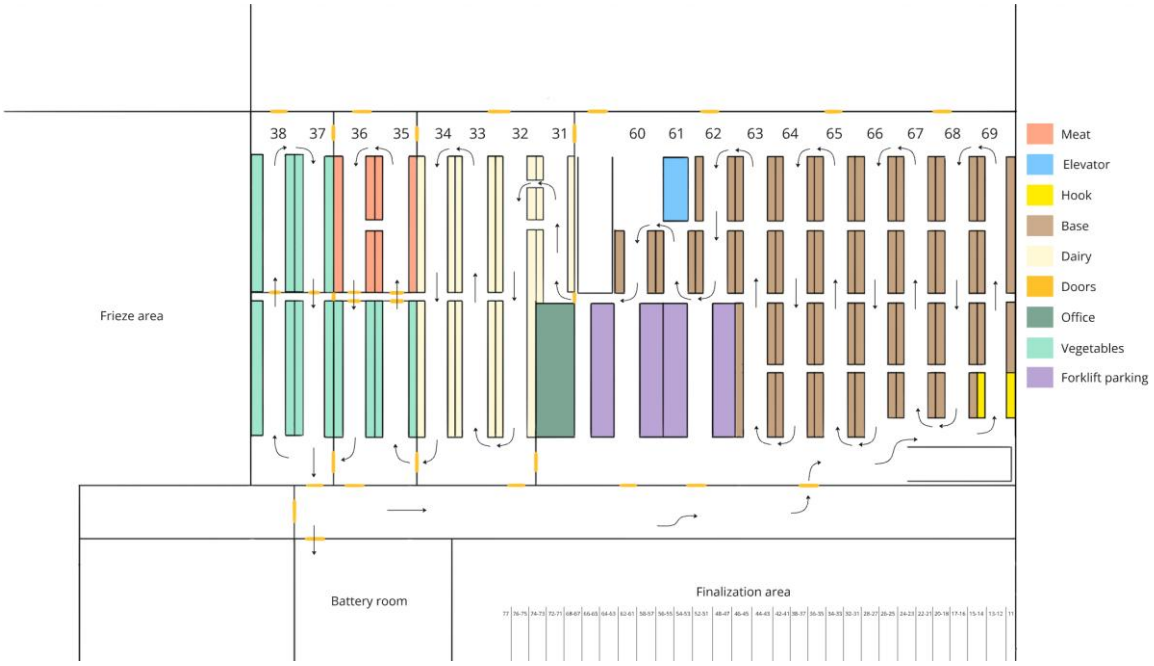


Figure 6: Work environment in the base area, the real size in base area is 5740 kvm.

Here is a user scenario with pick-by-voice, showing how it works and how the workers use the system. This user scenario could be longer, but the idea is to show what a workday might look like in a food warehouse. In the last eight slides of user scenario, showcase problems with the system and what can go wrong when picking goods. (see figure 7)

Hamayon Jafari, BFA3
hamayonjafaridesign@gmail.com

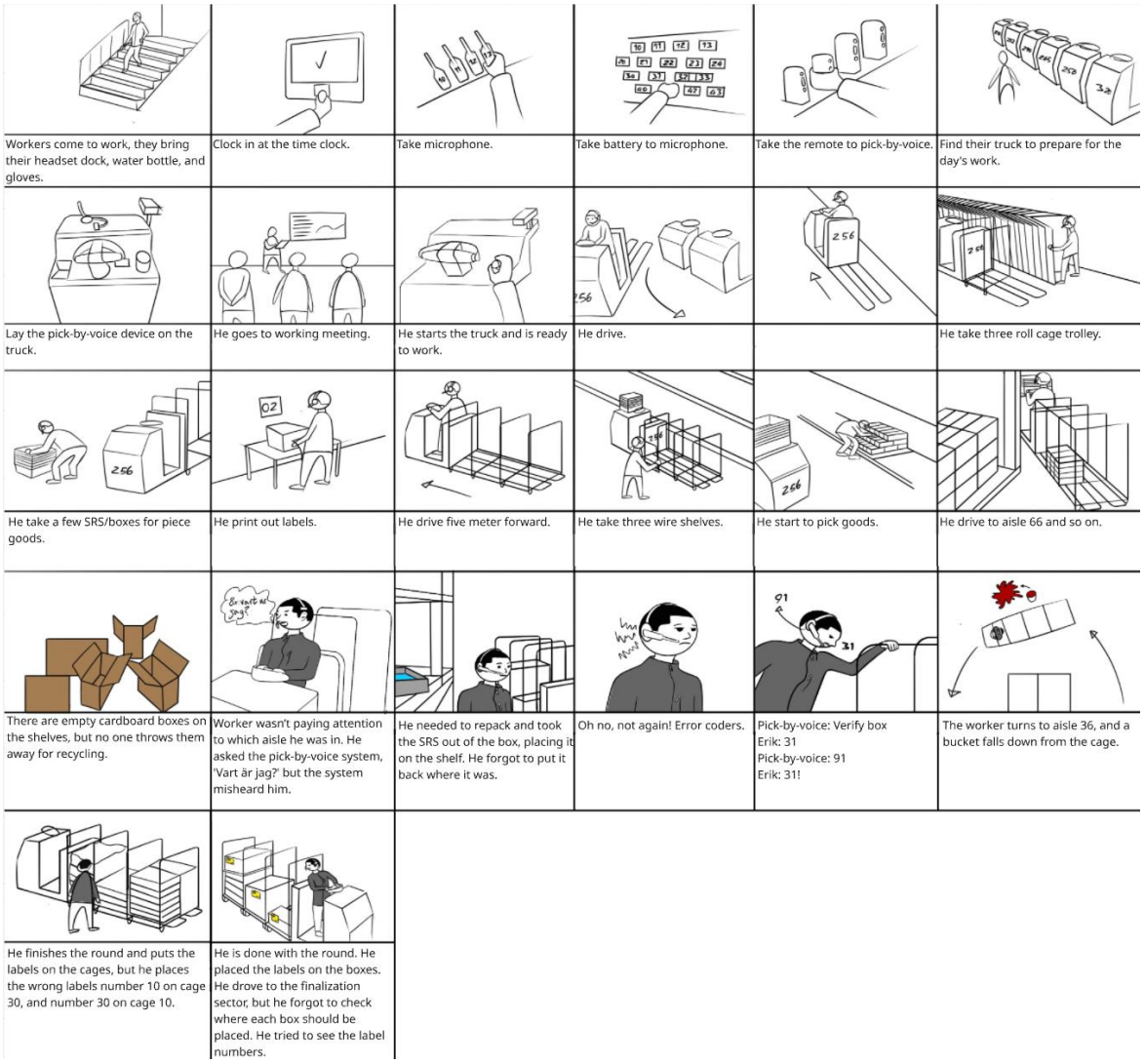


Figure 7: User scenario.

11.5. Activity analysis

Conducted an activity analysis to showcase the processes a worker goes through in their daily workflow and the problems they face. (see figure 8)

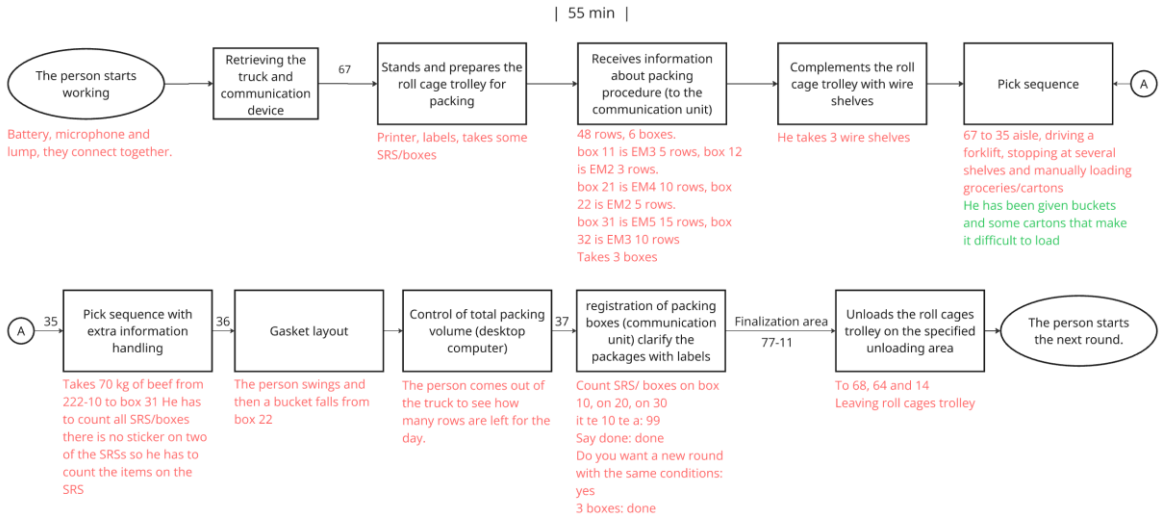


Figure 8: Activity analysis.

11.6. System map

I created a system map to illustrate the process that customer orders go through and how they are returned to the customer. This system map was made roughly because the company follows a privacy-focused business model, which prohibits walkthroughs in detail. (see figure 9)

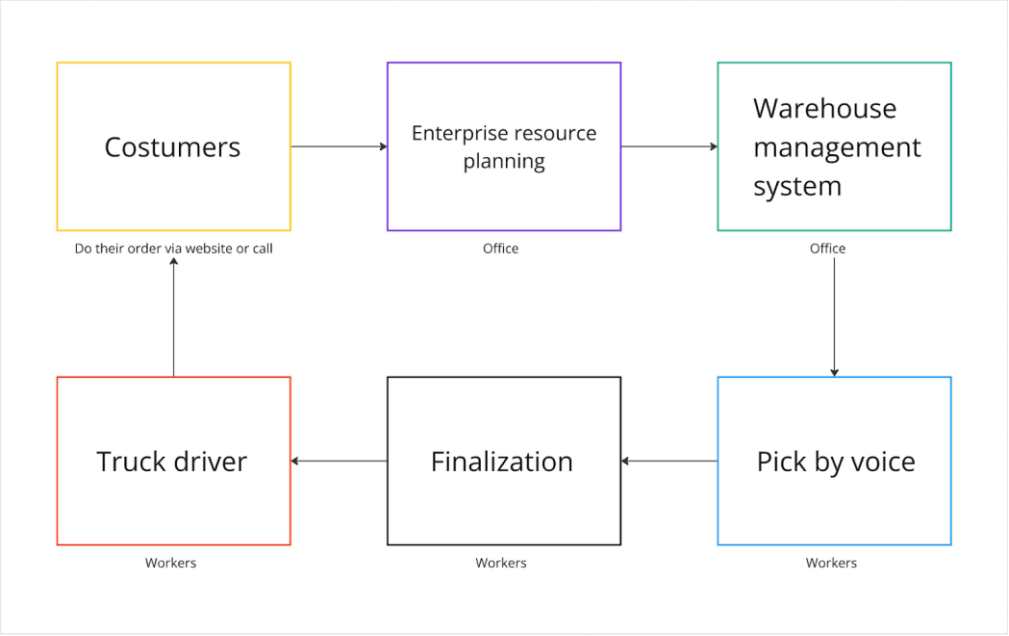


Figure 9: System map.

12. Process

The design process was carried out through research in different areas, user interviews, activity analysis, identifying problems with the system and the issues users face during a workday, creating a user scenario with the current pick-by-voice system to showcase the problems with it, and creating a system map to visualize the process from customer order to back delivery to customer.

12.1. Focus area

The figure below is an overview of how the thesis project took shape during the survey phase and what should be considered in the ideation phase. (see figure 10)

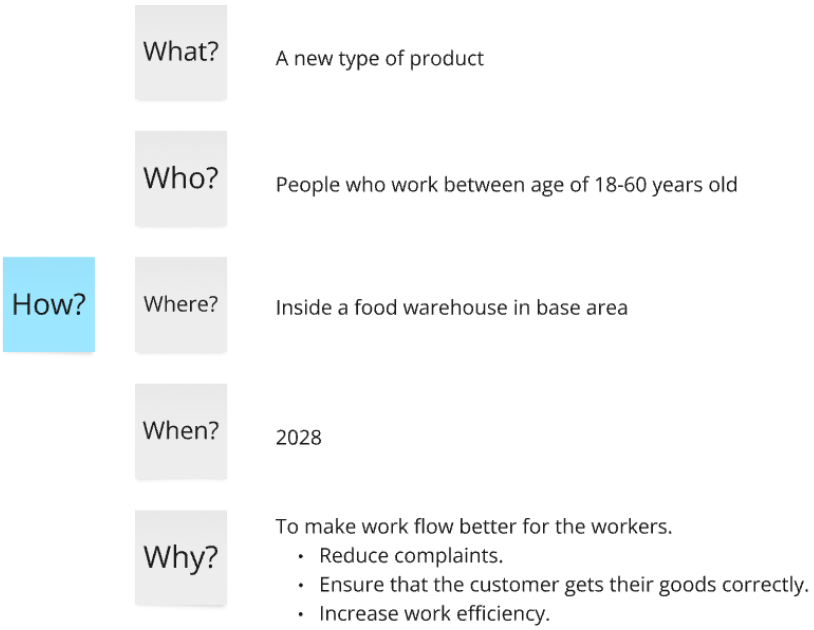


Figure 10: Focus area.

12.2. Function analysis

Based on research, interviews, and surveys conducted in the food warehouse, a function analysis was created. The purpose of the function analysis was to determine which functions are necessary. Throughout the project, the function analysis is continuously updated. (see figure 11)

HF = Huvudfunktion, SF = Stödfunktion, DF = Delfunktion

N = Nödvändigt, Ö = Önskvärt, IÖ = Icke önskvärt

Function	Words	Class	Priority	Remark
Facilitate	Pick item	HF	N	To the user
Improved	Connection	SF	N	To the user
Offer	Information	SF	N	To the user
Offer	Ergonomics	SF	N	To the user
Facilitate	Counting	DF	Ö	Goods
Simplify	Packning	SF	Ö	Roll cage trolley
Express	Security	DF	Ö	
Admit	Warning	SF	Ö	Roll cage trolley
Offer	Adjustability	DF	Ö	On the product
Facilitate	Cleaning	DF	Ö	On shelves
Simplify	Usability	DF	Ö	On the product
Minimize	Weight	SF	Ö	On the product

Figure 11: Function analysis.

12.3. Brainstorming phase

In this project, brainstorming was carried out after surveys had been done, and the problem area had been identified. The ideas that were sketched and written down were based on the research done for this project. A workshop was held with third year industrial design students from Sundsvall to present my project and develop the ideas I had sketched out during the brainstorming phase. Before the workshop began, the problem statement was explained, and the project was discussed to ensure that the participants had a clear understanding of it. (see figure 12)



Figure 12: Workshop with third year industrial design students from Sundsvall.

Hamayon Jafari, BFA3
hamayonjafaridesign@gmail.com

12.4. Overall concept ideas

Concept development: All ideas were analyzed after the workshop with the external supervisor from the warehouse, and then three concrete concepts were designed.

Concept 1

Designed a display that attaches to either the right or left forearm, depending on which side the user prefers. The aim is to allow the worker to pick goods and add them to the system by tapping on the screen, making the picking process easier and more efficient. The user can see all the information they need, and integrated speaker and microphone for the communication. (see figure 13)

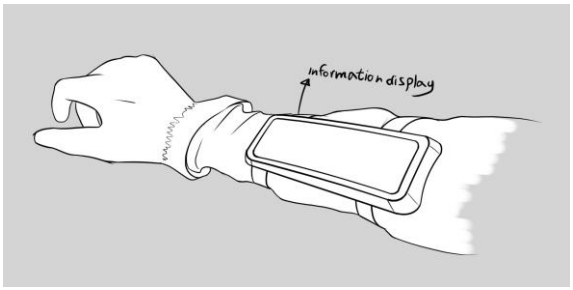


Figure 13: Concept 1.

Concept 2

Designed a product to be worn on jacket/sweater. It has a hook on the back to attach it securely, alternatively, there is a magnet that can be used when wearing a T-shirt or sweater. It features a camera and LiDAR sensor to verify the quantity of items picked and can recognize things in real time, providing feedback about the task. This concept comes together with pick-by-voice. (see figure 14)

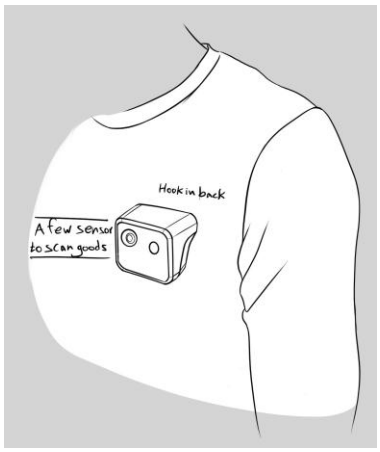


Figure 14: Concept 2.

Concept 3

Designed a headset and microphone integrated with smart glasses. The aim is to take up minimal space on the head to ensure comfort for the user. The product includes two sensors to track activities that are happening around the user. It features a single smart glass on one side, because some users wear glasses. This solution allows users to keep their own glasses while using the smart glasses as well. The glasses integrate a projector to display necessary information and provide visual help for packaging. (see figure 15)

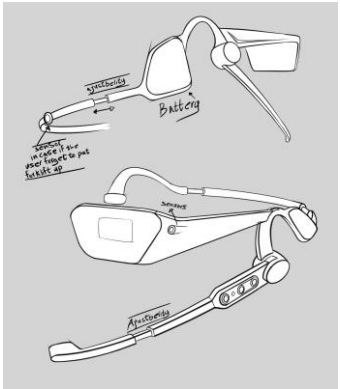


Figure 15: Concept 3.

12.5. Selection of concept

To be able to choose one of the concepts and go further in the project, I followed up with users to get their insights about the concepts, each concept was explained to the users to help them understand what it entails. This was done using four questions that participants had to answer about each concept:

- Which concept is reliable?
- Can the concept be used in warehouse work situations?
- Does the concept allow the user to be effective in their use case?
- Do you think this concept meets the requirements to solve the problem of today's warehouse work situation?

I asked seven people to choose one concept that I came up with. Two people chose concept 2, another two chose concept 3, and three chose concept 1. I concluded that concept 1 is the most relevant to proceed with based on user perspectives, however based on the problems I identified during the survey phase, I want to combine concept 1 with concept 2 to offer more features to address issues in today's warehouse work environment.

The three users who chose concept 1 did not like the idea of using pick-by-voice because they don't want to interact with a voice robot for an eight-hour workday and followed up to the interviews most user said that voice system mishear commend numbers, and the headset is seen as a disadvantage because it reduces social interaction in the workplace. It is important to create a better working environment to improve job satisfaction and reduce sick leave.

After following up with the user and showcasing the concepts, I compared and ranked them together. Concept 1 won with 70 points. (see figure 16)

		IDÉER & KONCEPT					
		Idé 1		Idé 2		Idé 3	
Urvalskriterier	**Viktning**	Poäng	Viktad poäng	Poäng	Viktad poäng	Poäng	Viktad poäng
Efficiency of use	3	5	**15**	3	**9**	4	**12**
Easy to understand	3	4	**12**	4	**12**	5	**15**
Ease of use	3	4	**12**	3	**9**	3	**9**
Ergonomics	3	3	**9**	2	**6**	3	**9**
Safer	2	5	**10**	4	**8**	3	**6**
Smart	2	4	**8**	3	**6**	5	**10**
Aesthetic	1	4	**4**	3	**3**	2	**2**
			70		**53**		**63**

Figure 16: ranking.

12.6. Form expression

To get inspiration and then define the design language of the final design proposal, the process began with the collection of images to build a form board that would shape the product and how it would look and feel. Five keywords were picked to describe how form and feel should be translated.

- Durable: The product should be durable so that in tough situations, like in the warehouse, it can be relied on.
- Sturdy: It should be sturdy in case you drop it, so that it wouldn't cause any damage.
- Robust: Where the feeling of security is formed in a combination of today's pick-by-voice products and future visions.
- Efficient: The interaction aspect should be efficient to maximize efficiency.
- Adaptable: The product should be adaptable in use case. (see figure 17)

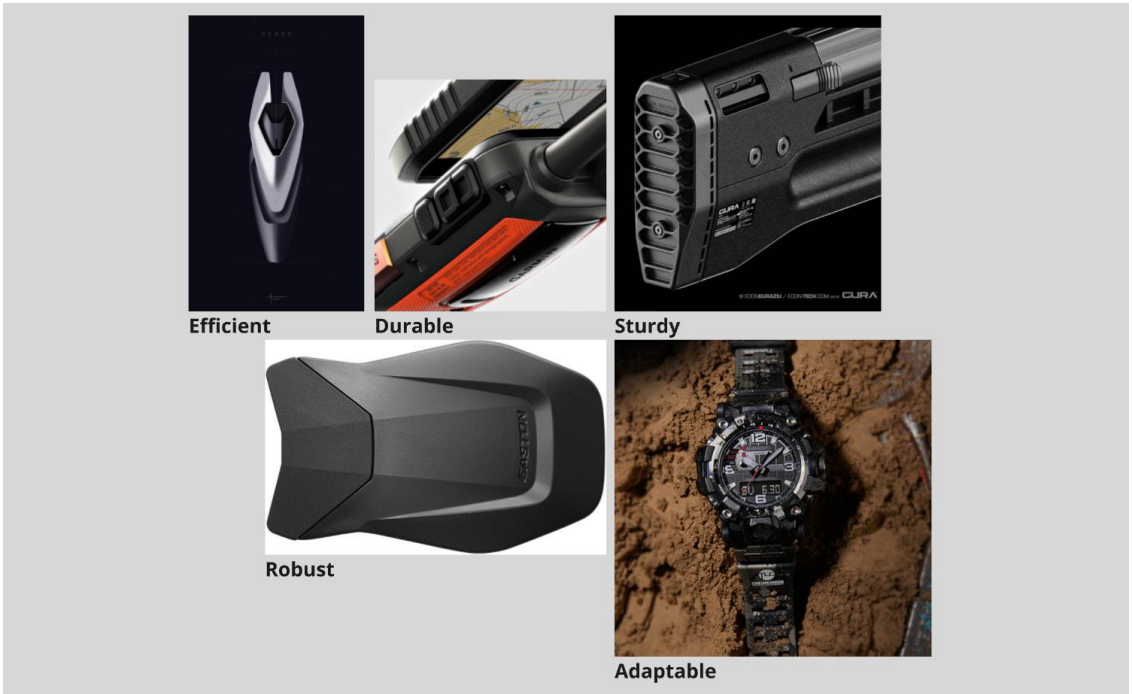


Figure 17: Form board.

12.7. Details about the chosen concept

The result would be two products connected to each other via Bluetooth. One of them is worn on a jacket or sweater. It is an AI-driven device that can track the task of the user, and it can recognize actions such as repacking and placing an (SRS) Standardized Return System on a shelf, and it notifies you to return it to the trolley by displaying a message on the other device's screen. It also has control

over activities for instance, if there is a problem during packaging, it can assist in improving the process. If you are walking down the street, this product can use its camera to identify objects and landmarks around you, it use this information to provide you with contextual information, like the name of the restaurant or the distance to the bus stop (Timesofindia, 2023).

There is an AI product that exist the have similar functionality, it is AI pin. It powered by a Qualcomm Snapdragen process and features a variety of sensors, camera, microphone, and accelerometer. It built-in projector that can be used to display information on the palm of your hand or other places.

The chosen concept involves having a display. Since workers in food warehouses are always using gloves, it is impossible to use a regular touch display. The display will be an industrial panel PC in small format. These panels are resistant to dust, extreme temperatures, shock, and vibration (Lavoie, 2024).

This product also includes a speaker and microphone for communication with managers or in case of accidents happened such as a bucket falling to the ground. It provides haptic feedback as well, so when the screen is tapped, the user receives confirmation through tactile response.

12.8. Size arm dimensions

To design the chosen concept, it is necessary to investigate forearm dimensions for both males and females. I checked this in the PeopleSize software, where you can measure different parts of the human body.

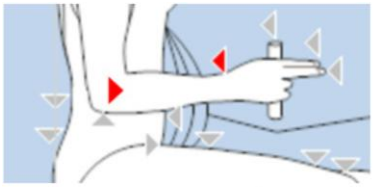


Figure 18: Forearm size. The size of these red arrows, like pointing, represents females aged 18 to 64 years old in Sweden. It covers between the 5% and 95% of the population, ranging from 155 mm to 281 mm.

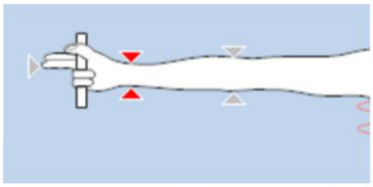


Figure 19: The wrist size for females ranges from 47 mm to 57 mm.

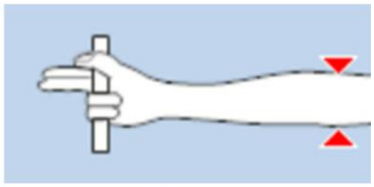


Figure 20: The size of these red arrows is 59 mm to 71 mm for females.

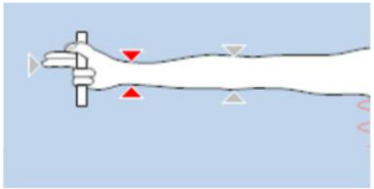


Figure 21: The wrist size for males between aged 18 to 64 years old ranges from 54 mm to 65mm.

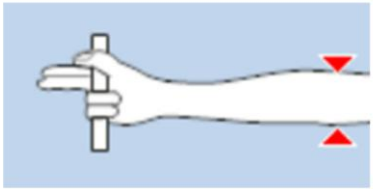


Figure 22: The size of these red arrows is 68 mm to 81 mm for males.

12.9. Design exploration of the chosen concept

Here is the overall ideation phase of the project, where I explored shapes for both the screen and the attachment product for jacket. (see figure 23)

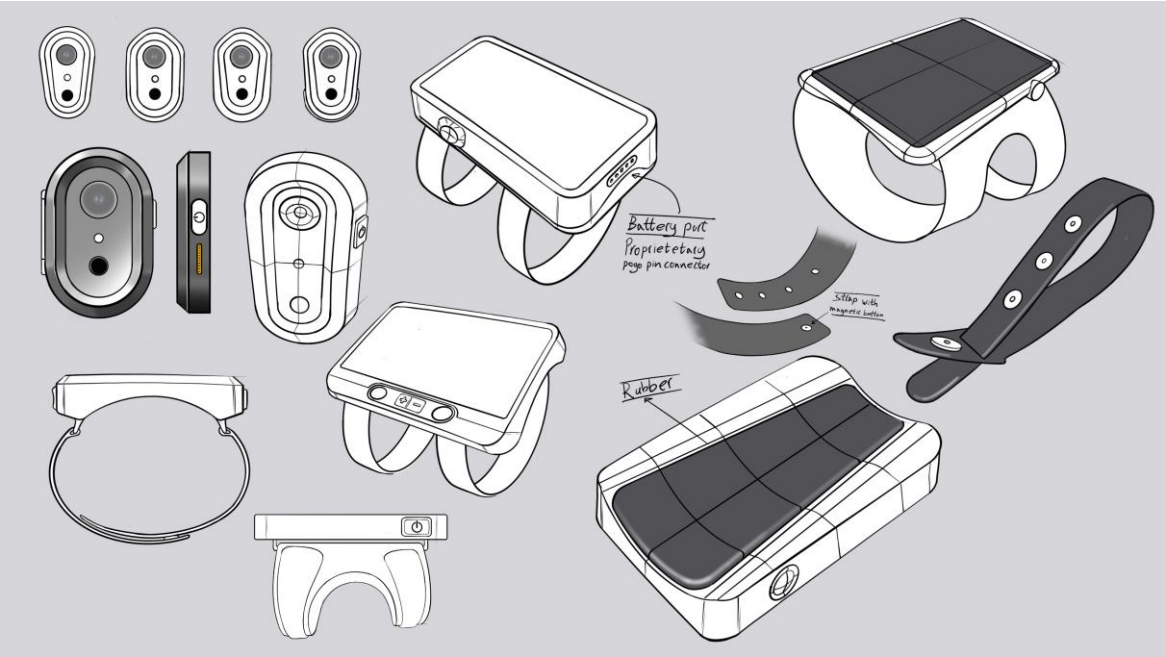


Figure 23: Ideation phase.

12.10. Prototypes

In this project, multiple prototypes were made to explore different sizes while considering forearm sizes for both males and females, these prototypes were tested on people with both small and large forearms. In the attachment header, you can find the dimensions for both products. At the warehouse I surveyed, I tested the product on their jacket and then tested the final product on the forearm, you can find it below. (see figure 24, 25 and 26)



Figure 24: Testing prototypes on people's forearms.



Figure 25: Prototypes from the early stage to the final one.



Figure 26: Testing prototypes on the human body.

12.11. Follow-up to user

The final concept was shown to three users; two of them were interviewed, the other one not and they said they liked it. They mentioned that this concept is much better than pick-by-voice and can greatly help in the picking process. With this concept, their wishes are fulfilled, and the necessary features are included.

Hamayon Jafari, BFA3
hamayonjafaridesign@gmail.com

12.12. Model development

To be able to develop the model, I designed it in CAD and printed it multiple times to refine the dimensions of the bottom, thickness, charging port, and so on. This is the modelling process from start to finish. (see figure 27)

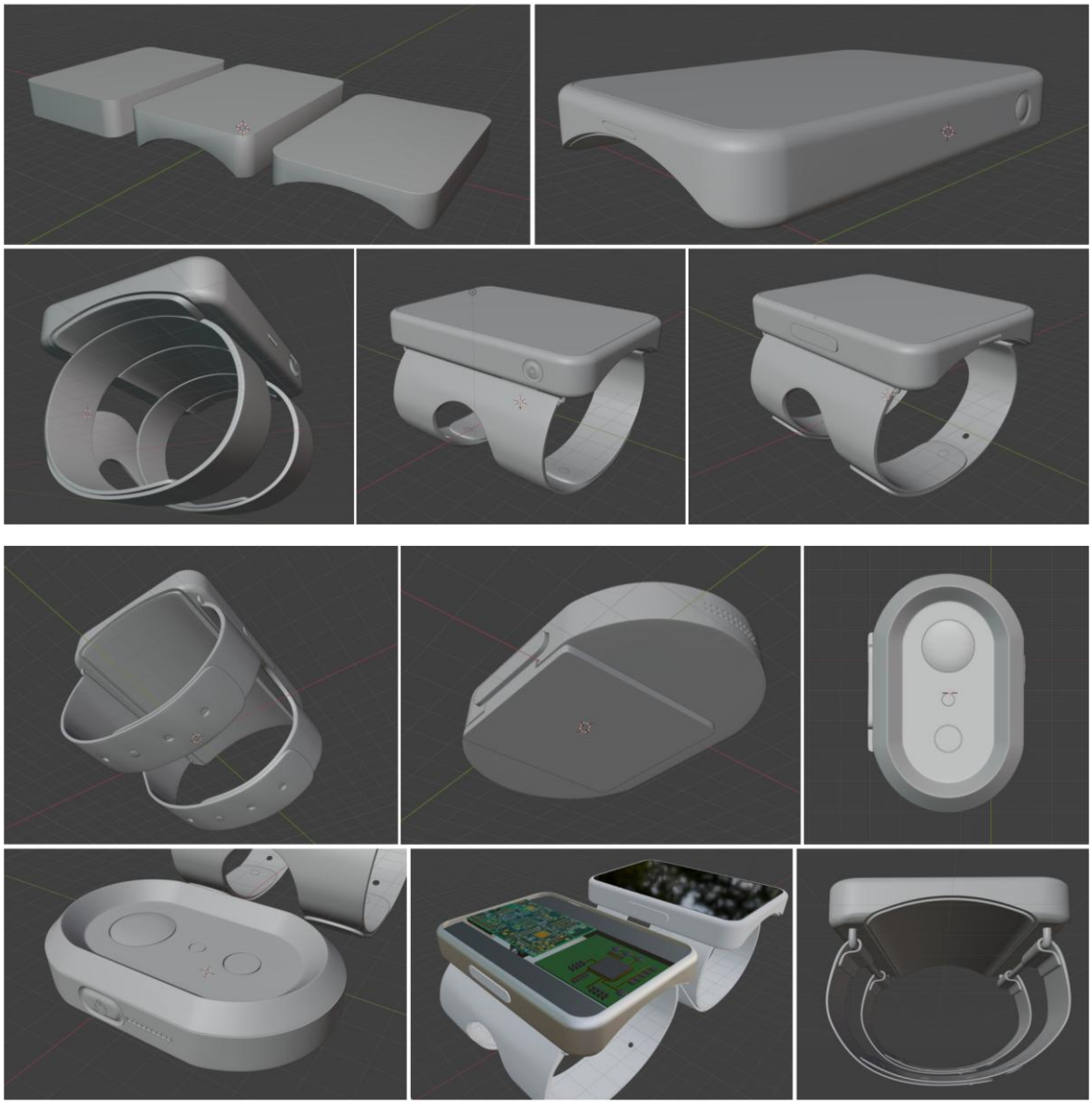


Figure 27: Modelling in Blender.

Hamayon Jafari, BFA3
hamayonjafaridesign@gmail.com

13. Results

13.1. Selection of materials and colour

For the housing of the product, I will use Acrylonitrile Butadiene Styrene (ABS) because it is lightweight, durable, and designed to resist drops, dust, and moisture. Although this material is not sustainable, there is no other alternative because it is resistant to cold weather which is beneficial for the battery life and not a mass-produced product. This material has a low production cost and is easy to produce by plastic manufacturers (Plastic Extrusion Technologies, 2021).

For the strap, it uses recycled PES material, and it is more durable than usual nylon webbing. For the button on strap, it uses the magnetic button. The display is made of durable glass that is resistant to impacts and drops. For the underside of the product that has a display, rubber will be used to provide comfort when held in the different size of forearm. The product is two-tone in colour: dark gray and dark blue. These colours were chosen because they are reliable and suitable for warehouse tools, I choose two colours because I wanted the product to have a distinctive feeling. (see figure 28)



figure 28: colour codes.

13.2. Scenario

Here is a user scenario that describes the problems with the current system (pick-by-voice) and with my design proposal. These are a few examples of problems that the new system can solve. (see figure 29)

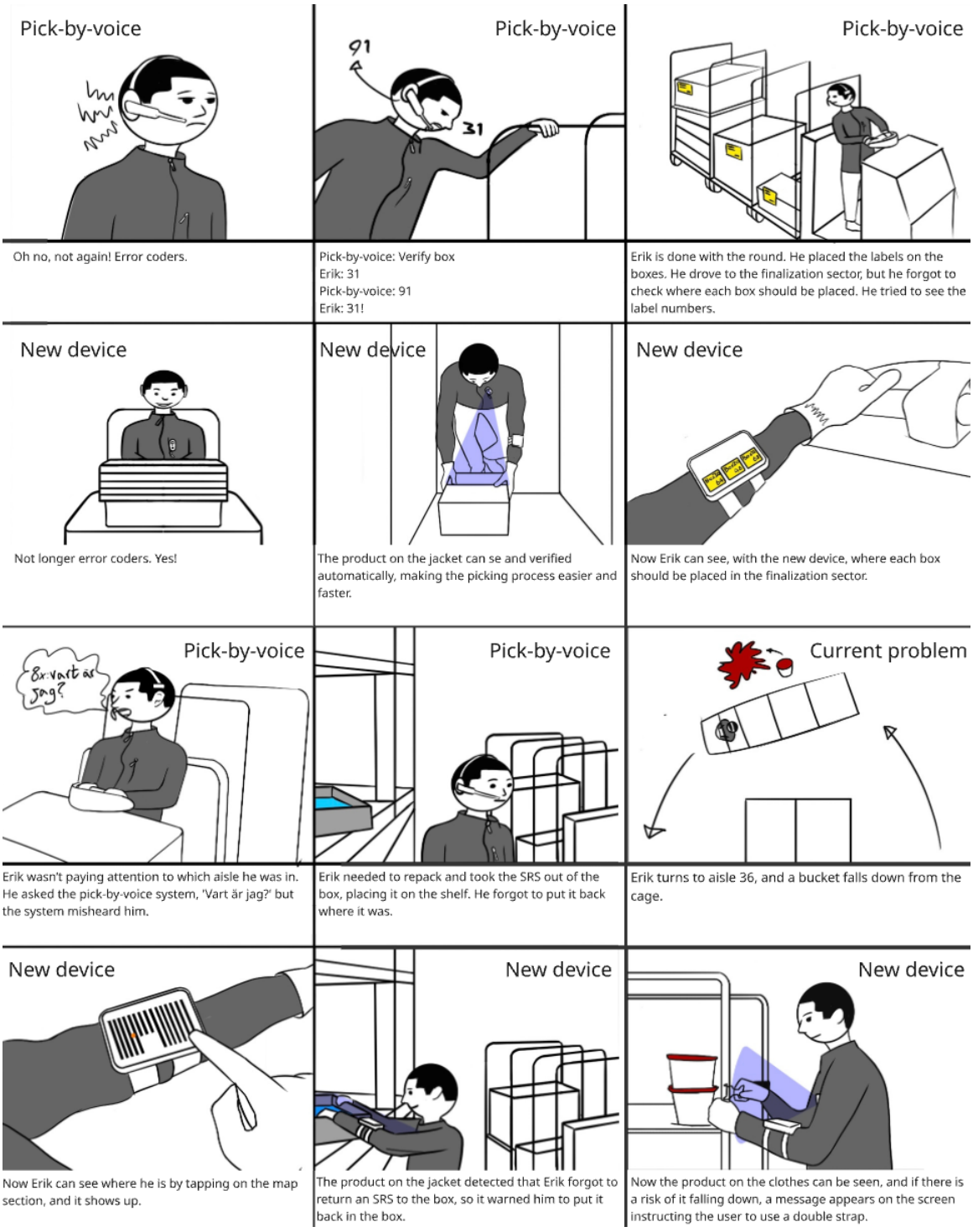


Figure 29: User scenario.

13.3. Interface distributed

This distributed interface provides an overview of how it works. It includes approximately 90% of the features. (see figure 30 and 31)

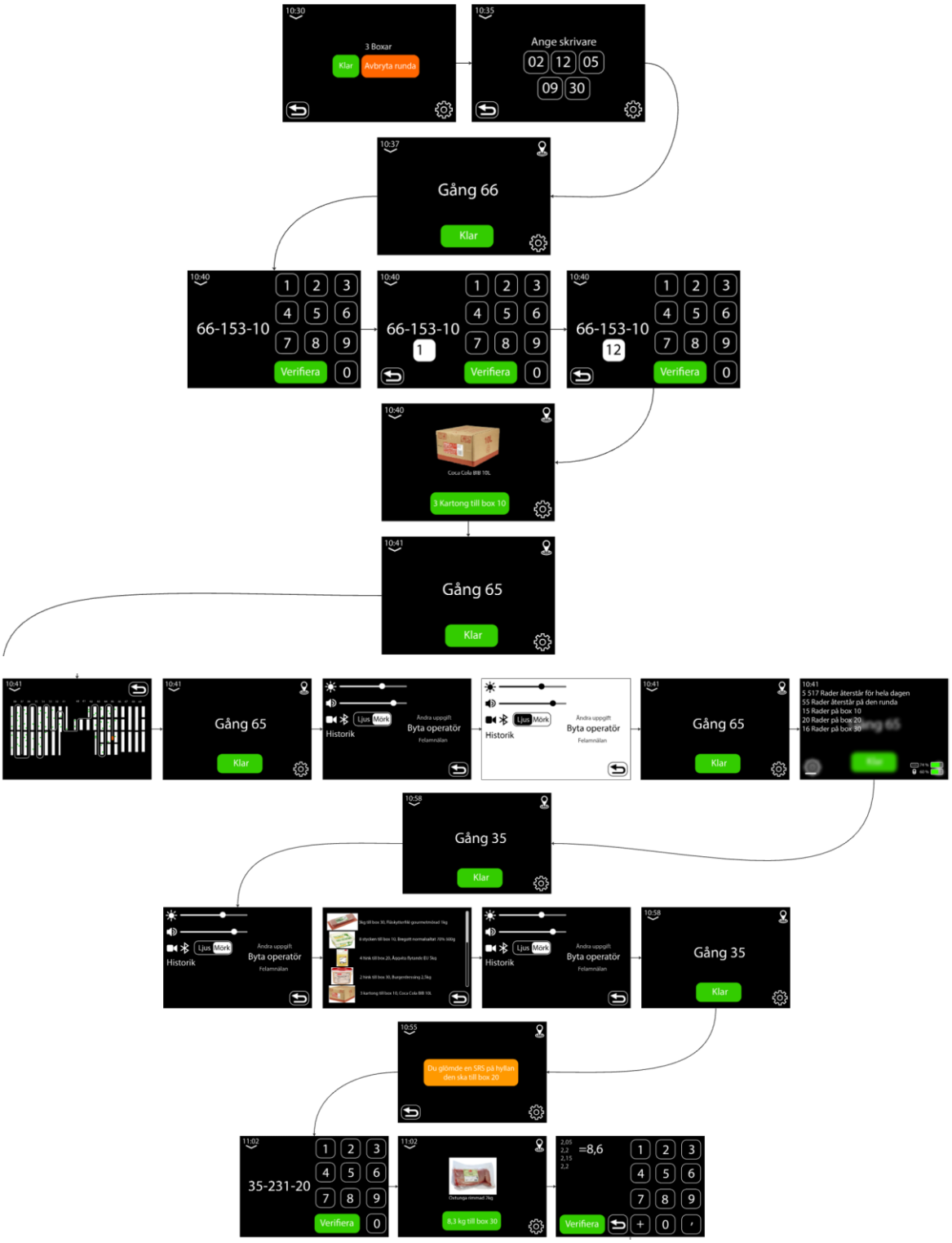


Figure 30: Interface, see the continuation of the image in the next page.

Hamayon Jafari, BFA3
hamayonjafaridesign@gmail.com

UMEÅ UNIVERSITET

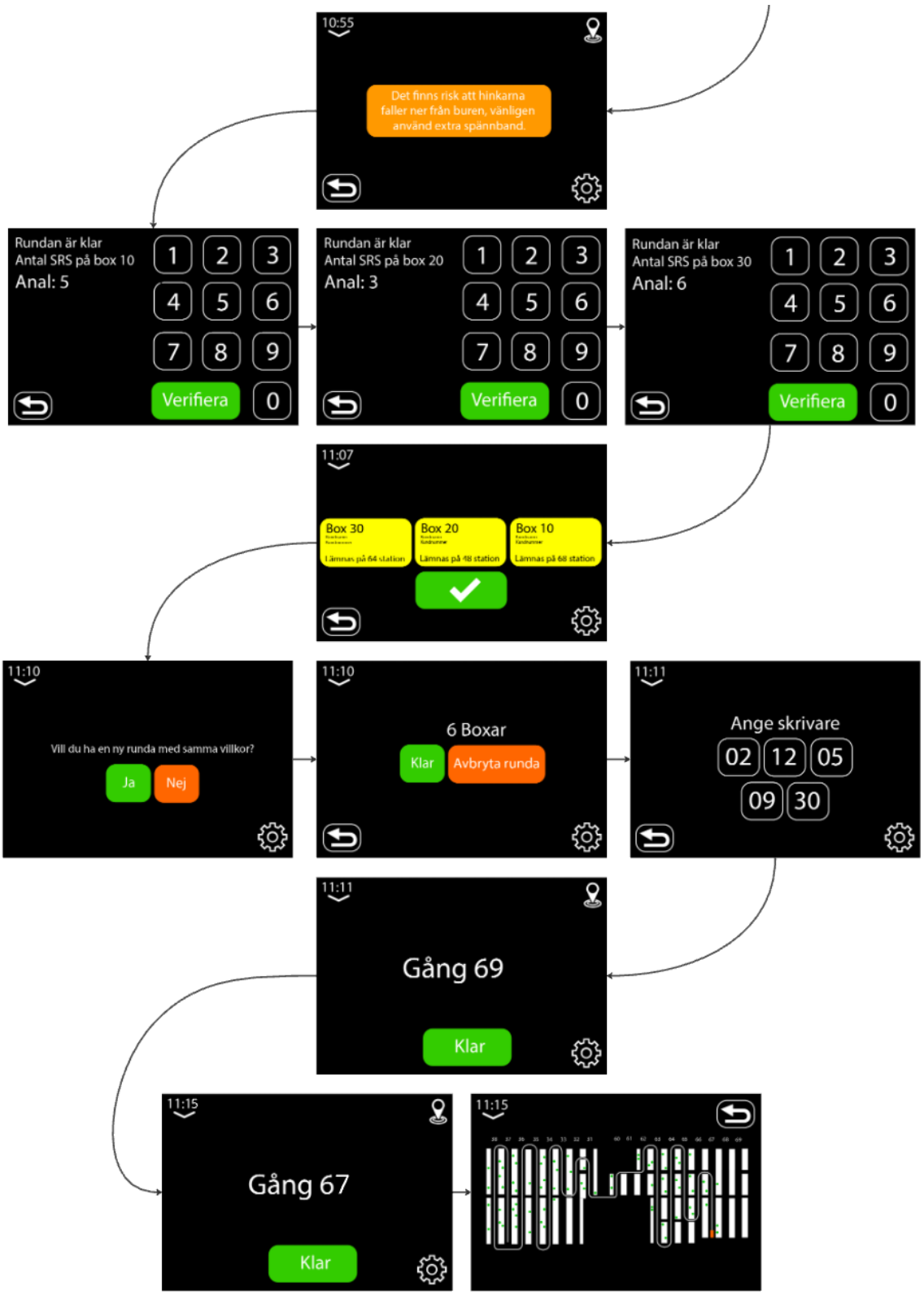


Figure 31: Interface.

13.4. User scenario with interface

Here is a complete process of how a user scenario, with interfaces showing every step of using the new product with better functionality compared to the current version “pick-by-voice”. During the picking process, the user wears the two devices on themselves: the one with a camera is worn on the chest area, and the other one with a screen is worn on the forearm. I did not show the entire picking order because it would have been too much to include. (see figure 32 and 33)



Figure 32: User scenario with interface, see the continuation of the image in the next page.

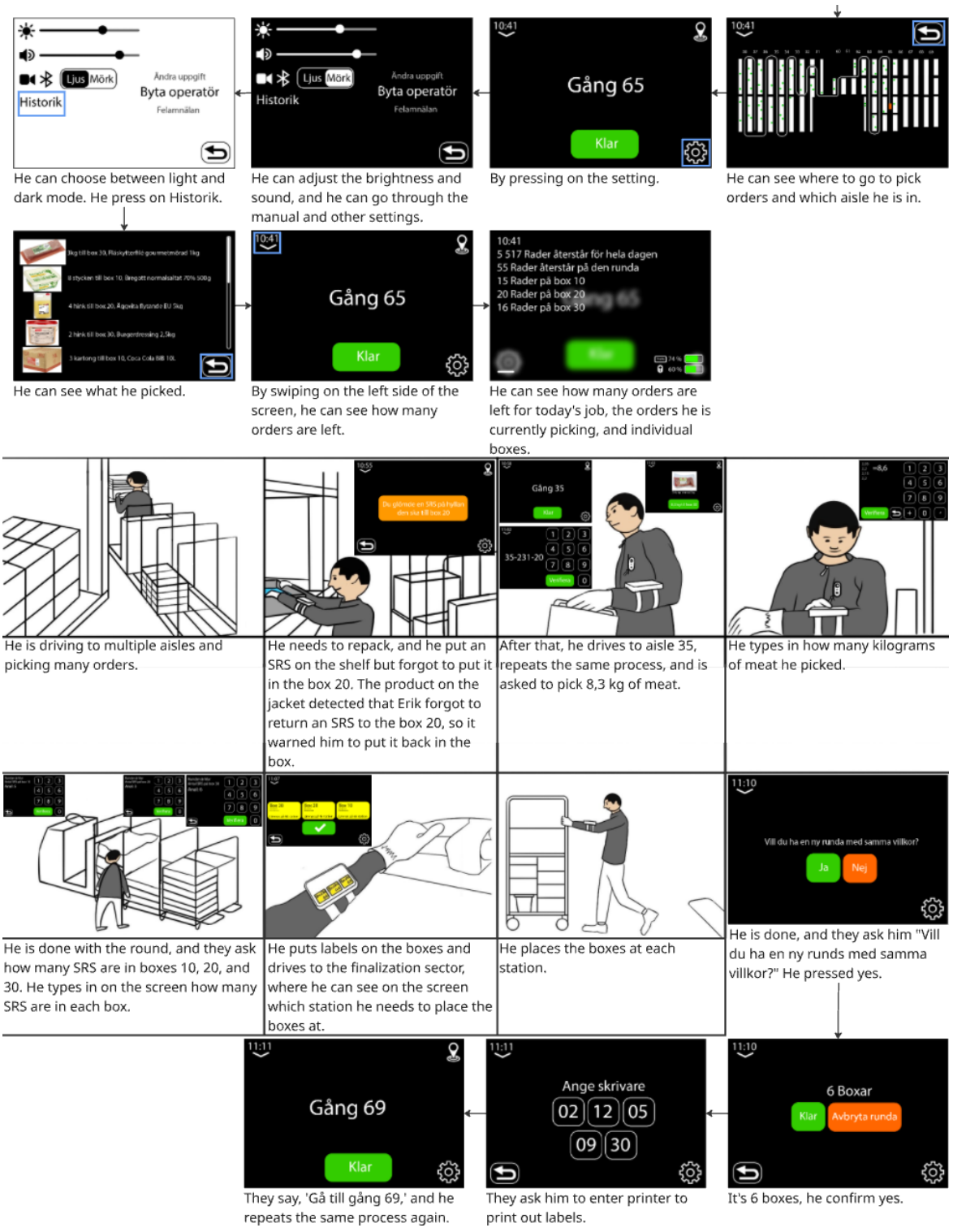


Figure 33: User scenario with interface.

13.5. Product name

At the end of the project, I gave the product a name. 'Handy Picking' is the final name proposal because it represents a practical system that streamlines picking process in the food warehouse.

13.6. Renderings

The way to connect two devices is to first turn them on by pressing the power button. Then, place the Bluetooth connection ports close to each other. After a few seconds, the devices will load and confirm that they are connected. Once connected, log in, select the correct operator, and go through a few manuals before starting your first round of picking. (see figure 34)



Figure 34: Connection.

There are two ways to attach the product to a jacket. The first is to slide a hook into the designated slot and then place it into the pocket (see figure 23). The second method uses an attachable magnet to connect the parts together, which is intended for use when the user is wearing a sweater. (see figure 35)



Figure 35: The way to wear the product.

Here is an exploded view of the product that you can wear on your forearm. (see figure 36)

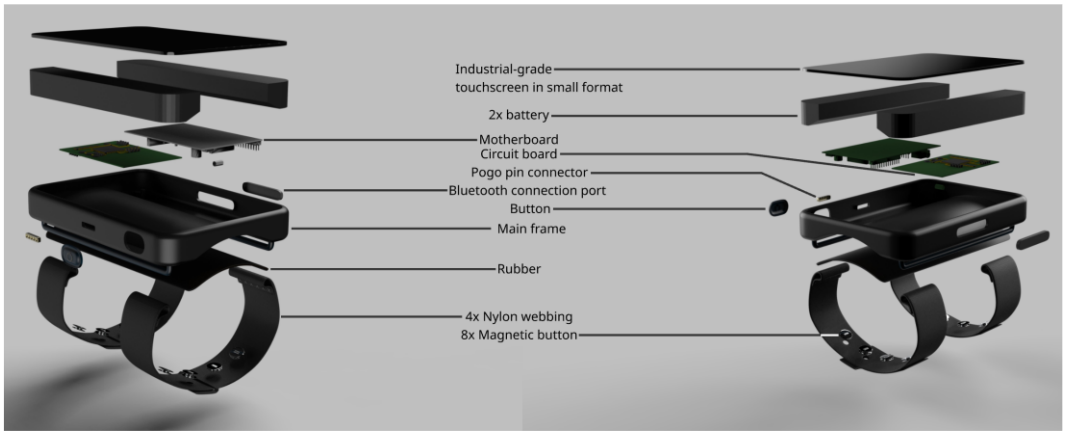


Figure 36: exploded view.

Here is another exploded view of the product that you can wear on your jacket. (see figure 37)

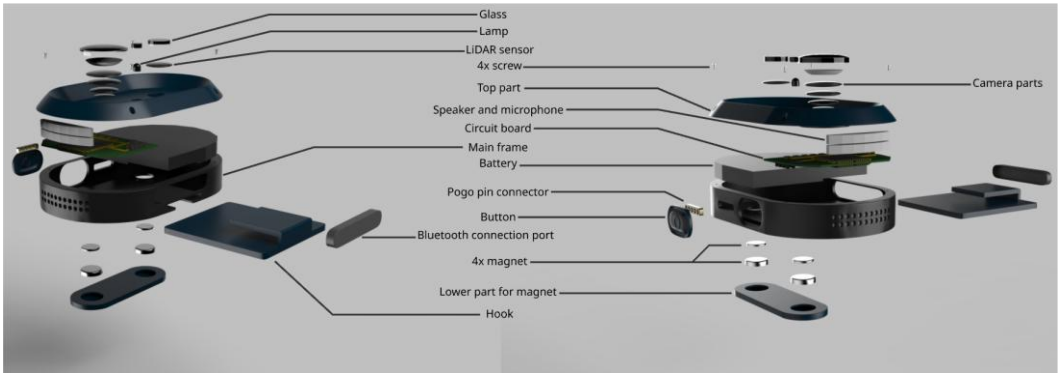


Figure 37: exploded view.

An overview of the product that you can wear on your jacket. (see figure 38)



Figure 38: Overview.

An overview of the product that you can wear on your forearm. (see figure 39)





Figure 39: Overview.

14. Follow-up on the UN's Global Goals

This project aims to address two of the United Nations' Global Goals: Goal 8, Decent Work and Economic Growth, and Goal 9, Industry, Innovation, and Infrastructure. Goal 8 focuses on ensuring safe, fair, and dignified working conditions, especially for women and adults. It also aims to reduce unemployment. Through my design proposal, the hope is to reduce picking mistakes—benefiting the warehouse, its employees, and its customers. Additionally, to supports small enterprises by encouraging financial support to help them develop and produce products that improve the efficiency of other businesses like in the food warehouse industry.

Goal number 9 can contribute by encouraging supply companies to create inventory management software that reduces errors and guides pickers to items more quickly and accurately. The aim was to design a product to assist humans in picking process, not to replace their jobs by robot or AI driven product. This aligns with UN's global goal number 9, which is to create innovations that support humanity, not work against it.

15. Discussion

The project presents a future concept involving two wearable devices for workers who pick goods from shelves and drive low-level order pickers through different aisles to the finalization sector. The hope is that this solution will improve the workflow in the picking process in the future and increase user satisfaction.

During the thesis project, the timeline was based on a schedule created before the project began. It was followed for the most part, although the direction of the project naturally shifted do to the based-on user perspectives and their input. I did not have permission to include images from the food warehouse, but I did my best to describe what a warehouse might look like under section “11.4. Work Environment,” and to illustrate what the picking process could involve.

This project taught me, as tested design methods were used to drive the process forward and create a tangible, innovative product. Although the outcome resulted in something that already exists “a wearable computer”, the design was guided by user needs. This version of the wearable computer is smaller and thinner, tailored for the working conditions, as workers often need to reach into tight spaces when picking items from a roll cage trolley. Therefore, a compact device is necessary one that can fit easily into confined areas.

If handy picking were to be launched in the year 2028, it would be relevant because advanced technology will have improved, and there would be enough time to develop and test it in real-world use cases before going into production.

A counterargument is that the result is not the most optimal because what if some features don’t work for a specific task? For instance, if the software loads like a phone, because sometimes phones loading, which can be annoying for the worker. It can also happen that you lose the attachable magnet on the product you're wearing on your sweater, or the hook on the same product. These are factors that need to be considered.

In conclusion, I wished I had more time to explore the topic further. However, I am happy with the result because when I showed it to the users, they were satisfied.

16. Reference list

Figure 1: British Design Council (2005). The Double Diamond. Design Council. https://www.designcouncil.org.uk/our-resources/the-double-diamond/

Figure 2: United Nations (2023) Available at: https://sdgs.un.org/goals

Figure 4: Del Buono, A. (2015). Voice-picking technology continues to grow, innovate: Bev industry. https://www.bevindustry.com/articles/88824-voice-picking-technology-continues-to-grow-innovate

Honeywell (n.d.). Empower Your Mobile Workforce. Available at: https://automation.honeywell.com/gb/en/software/productivity-solutions/order-picking-fulfillment/voice-picking (Accessed: 2025-03-30).

Zebra Technologies (n.d.). HS3100 Rugged Headset Series. Available at: https://qac-downloads.zebra.com/us/en/products/mobile-computers/wearable-computers/hs3100.html (Accessed: 2025-03-30).

Körber (n.d.). Voice Solutions. Available at: https://www.koerber-supplychain-software.com/en/supply-chain-solutions/voice-solutions (Accessed: 2025-03-30).

Real Time Logistics (n.d.). Voice Picking Systems. Available at: https://www.realtimelogistics.com.au/technologies/voice-picking-systems/ (Accessed: 2025-03-30).

Logistics IT (2014). Voiteq's VoiceMan picking solution now available in French. Available at: https://www.logisticsit.com/articles/2014/10/23/voiteqs-voiceman-picking-solution-now-available-in-french/ (Accessed: 2025-03-30).

Lucas Systems (n.d.). Advancing Warehouse Management with Voice Picking Software. Available at: https://www.lucasware.com/voice-directed-warehousing/ (Accessed: 2025-03-30).

Jenkins, A. (2021). What Is Voice Picking? How It Works, Benefits & FAQs: Net Suite. Oracle Net Suite. https://www.netsuite.com/portal/resource/articles/inventory-management/voice-picking.shtml

Udikas, M. (2019). Norska mejeriet Tine provar plock med smarta glasögon: Transport nytt. https://transportnytt.se/norska-mejeriet-tine-provar-plock-med-smarta-glasoegon/

Mecalux (n.d.). Pick by voice technology is on trend once again. Available at: https://www.mecalux.com/blog/pick-by-voice#:~:text=Pick%20by%20voice%20gains%20new%20momentum,-Increased%20efficiency%20in&text=A%20study%20by%20consulting%20firm,over%20%243%20billion%20in%202025. (Accessed: 2025-03-30).

Haslam, C. (2024). Review: Even Realities G1 Smart Glasses: Wired. https://www.wired.com/review/review-even-realities-g1-smart-glasses/

Lavoie. (2024). 7 Step Process to Selecting an Industrial Panel PC. Onlogic https://www.onlogic.com/eu/blog/the-7-things-to-consider-before-buying-industrial-panel-pc/

Plastic Extrusion Technologies (2021). What is ABS Material? Available at: https://plasticextrusiontech.net/what-is-abs-material/ (Accessed: 2025-03-30).

Figure 37: Svenska Retursystem (2023) Available at: https://www.retursystem.se/sv/vart-retursystem/srs-lador

Timesofindia (2023). Explained: What is Humane AI Pin, how it works and what it does. Available at: https://timesofindia.indiatimes.com/gadgets-news/explained-what-is-humane-ai-pin-how-it-works-and-what-it-does/articleshow/105138840.cms (Accessed: 2023-11-11).

17. Attachments

Ten interview questions with answers in a summarized format.

1. How does it feel to use pick-by-voice?

The advantages are that it is easy to pick goods and keep pace with the work.

More efficient than using paper with an order list. When it works well, it makes the job easier.

The disadvantages are that the voice recognition is flawed and often hears error codes. Annoying to have to repeat yourself or correct errors afterward. The system feels old, lags often, and lacks updates. Error codes and technical problems make some days very tough to work. The pick-by-voice lacks flexibility. Sometimes, when you pick goods, the system mishears the quantity you picked. When that happens, it is difficult to undo or correct mistakes without assistance from the warehouse office to correct there in the system.

The pick-by-voice is a good idea in theory but does not always work optimally in practice. Voice recognition is a major source of frustration, and technical problems make the workday more difficult. At the same time, it is seen as a more efficient solution than paper lists, but there is much that can be improved, especially when it comes to the modernity and flexibility of the system.

2. Is there anything about it that is annoying?

The pick-by-voice lags, freezes, or does not hear certain numbers and commands, which creates frustration and negatively affects the workflow. Specific numbers such as 1, 2, 3, 8, and 9 seem to be particularly problematic, as the pick-by-voice does not always register them correctly. It is also mentioned that some commands are not understood, and sometimes the pick-by-voice is completely silent for several minutes. Error codes are experienced as annoying, and in some cases, to prevent of error codes they changed the pick-by-voice to "black version of pick-by-voice" before the system finally gives an error code. The current version of pick-by-voice that is used in the warehouse is blue. This leads to users having to repeat their commands several times and wait for the error to resolve, which can take up to five minutes.

3. Have you used virtual reality (VR) before, or are you familiar with it?

Five of the workers had not used “VR” previously. The interviewees think that VR at work would be an interesting and fun solution for the future. The same person wonders if it could work via an app and sees advantages in using VR when picking goods, especially if the information is presented clearly.

Another person has previous experience with both "pick-by-light" and “pick-by-voice” systems. He prefers pick-by-light. His opinion is that the pick-by-voice did not work as well as the pick-by-light and created stress. In addition, the headset is seen as a disadvantage, as it reduces social interaction in the workplace. It is emphasized that a pleasant working climate is important, both for job satisfaction and to reduce sick leave. One of the interviewees also expressed that work should be fun and that they are tired of pick-by-voice.

4. Does the pick-by-voice often get error messages? When it does, what do you do in that situation?

In general, the number of error codes has decreased compared to before, but they still occur and can be frustrating. There are two types of error codes:

Occurs when the connection with the Warehouse Management System (WMS) is lost but returns to normal operation after reconnecting. System silence "ghosting" - the pick-by-voice stops responding completely, probably due to a connection problem.

Previously, the problems were more extensive, but measures have been taken, and the pick-by-voice is now working better. The impact of the error code varies depending on the work area; for example, more error codes occur in the cold area and on specific aisles such as 36, 37, and 38. Adjustments have been made, such as moving equipment, which has improved the situation. Some tools, such as different lumps (blue and black), can also affect the error rate, where the black lump seems to have worked better for some users.

5. Do you have any wishes for what the pick-by-voice could have? It could be anything.

- Ability to see the number of rows left both in the entire warehouse for today's job, the round you are running and in a specific box.
- That pick-by-voice shows how many rows are left in the round before verification.
- Improved system connection to reduce errors and interruptions.

- Built-in clock function in the pick-by-voice instead of having to check the mobile phone.
- Advanced information about upcoming goods, for example, knowing that the next item is one or many soft drinks.
- When entering weight, the pick-by-voice should immediately hear the exact weight to facilitate handling.
- Calculator function in pick-by-voice to automatically add weights.
- Improved handling of double rows, so that the process is smoother and faster.

Need for a command to be able to backtrack and verify the correct goods instead of just seeing the history. One of the users that I interviewed said that sometimes he picks goods in the pallet instead of the roll cage trolley. He said it is important to pick in a way that the pallet doesn't not get pyramid-shaped, the weight is needed to create equilibrium.

The desire is to eliminate error codes and for the pick-by-voice to function flawlessly. In summary, more precision, predictability, and clarity are requested in the pick-by-voice, with better connection, voice recognition, and information functions.

6. Is it difficult for the pick-by-voice to hear what you say?

Some numbers, especially "31", are difficult to hear for the pick-by-voice - it is often misinterpreted as "91" or something else. The pick-by-voice sometimes guesses commands if it does not hear the entire phrase correctly.

Examples of common misinterpretations:

- "4" is heard as "7"
- "1" can be interpreted as any number
- One of the users says "Tea" instead of "3" and the system counts as 3.
- "3" is heard as "9"
- "Check number 44" can be heard as "talk man" and roll cage trolley 22 can also be misinterpreted, or 74 can hear "on the fill".

When does it work better? The pick-by-voice works better when picking cartons, but worse when handling vegetables and weighed goods. Previously, the problems were worse, but some improvements have been made.

Desired improvements:

Better voice recognition, especially for difficult numbers. Fewer misinterpretations, so that the pick-by-voice does not guess incorrectly. Remove error codes so that the picking process is smoother. In conclusion, voice control is a big challenge; many users want more accurate and reliable voice recognition.

7. What is the most annoying thing about pick-by-voice?

Three of the users said that nothing was annoying about the pick-by-voice, while five users found it annoying to use. Pick-by-voice sometimes disconnects, which disrupts the workflow. It is slow at times, and some commands are too long – a shorter option would have saved time. The system takes time to respond, which can be stressful, especially when there is a queue in the aisles. The headset and dook are uncomfortable to work with, and there are two types of headsets dook: one has big ear cushions, which are comfortable for the ear, and the other one is smaller, which is uncomfortable, and one of the users said a small microphone would have been more convenient.

What the user wants is desired improvements, such as a faster response and better connection to avoid disconnects. Shorter commands so that you don't have to listen to unnecessary information. Lighter and smaller headsets take up less space and are more comfortable to use while working. Better voice recognition, so that the pick-by-voice hears what the user is saying. In summary, users want a faster, more reliable, and more convenient product with better voice control and a more ergonomic headset.

8. Do you have problems counting goods, especially when there are a lot of them? (see figure 40 about SRS)

One of the users said that the frustration is that if there are SRS/boxes to pick and the stickers are not on any of them and then pick-by-voice says "pick 100kg," each box must be weighed individually. It's a bit frustrating, but otherwise, if it is possible to count. An old person said that mental arithmetic is not my strongest skill, so I have a regular calculator to count.

For instance, pick 100kg banana cartons. It's a bit difficult to know how many cartons you need to pick because each carton weighs 18.5kg, and then you must think of four cartons and then a bit more. In the meat section, one of the users was going to take salmon, and there was no sticker on the SRSs/boxes, so he had to count all the kilos with his mobile phone. It works, but it takes time.



Figure 40: 6 different types of SRS, retursystem.se (2023).

9. Do you listen to music when picking goods?

Three people do not listen to music, and seven people listen to music with a speaker.

10. How do you handle situations where you need to repack the box?

Stability is important – try to stack the goods as stably as possible, for example by placing small buckets on SRS/box. Most people repack when it is necessary, but some try to avoid it by having a good packing strategy from the beginning. If it is necessary, repack quickly and preferably put goods on the floor instead of putting them on the shelf. Grid shelves are often used to protect sensitive goods and place heavier goods at the bottom, which reduces the need for repacking.

One of the users finds that picking things with the roll cage trolley is challenging because there is limited working space. Strategies to avoid problems include not repacking in tight areas where there are people in the aisles with forklifts, but instead, driving a little forward and repacking having the goods close to the roll cage trolley that you are packing. Users strive for stable packing and efficient working methods to minimize repacking. Grid shelves are used to simplify the work, while some prefer to avoid repacking entirely by having a well-thought-out picking strategy.

Dimension for the product (see figure 41)

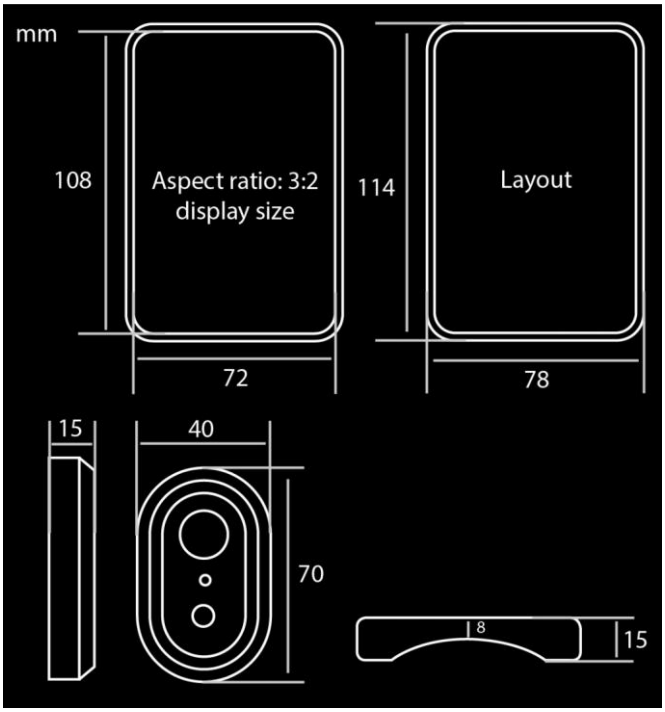


Figure 41: Dimension size.

Screen prototyping and deciding the dimensions of the calculator buttons, as well as determining the appropriate screen size. It looks like that the gloves used in the base sector of a food warehouse are typically not as thick as the gloves used in the frozen sector. (see figure 42)



Figure 42: Screen prototyping and gloves.

Hamayon Jafari, BFA3
hamayonjafaridesign@gmail.com

I did research on VR technology for Concept 3 (see the 'Overall Concept Ideas' heading) and categorized it into three sections.

